

Valence Alignment Comparison: Human-AI vs Human-Human vs AI-AI

Cross-Condition Contrasts on the Exchange-Aligned Data Model

PENPAL Project

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This notebook compares **valence alignment** across the three experimental conditions using the new exchange-aligned structure.

- **Human-AI**: Human co-writing with an LLM
- **Human-Human**: Two humans co-writing together
- **AI-AI**: Two AI agents co-writing together

Important structural note:

- `author_1` / `author_2` are now treated as **slot 1** / **slot 2** in the cleaned exchange sequence.
- In **Human-AI**, those slots also map onto human vs AI roles.
- In **Human-Human** and **AI-AI**, the slots are positional sides of the exchange, not stable identity labels.

The analysis window is restricted to **complete exchanges** and the default aligned turn range (analysis_turn 1 to 9).

Primary alignment object. Each story contributes one unsigned, label-invariant dyad asymmetry magnitude:

$$A_{\text{valence}}(i) = |\text{atanh}(r_{\text{first} \rightarrow \text{second}, i}) - \text{atanh}(r_{\text{second} \rightarrow \text{first}, i})|$$

where $r_{\text{first} \rightarrow \text{second}, i}$ is the per-story Pearson correlation between same-row chronological first and second turns, and $r_{\text{second} \rightarrow \text{first}, i}$ is the correlation between the previous row’s chronological second turn and the current row’s chronological first turn. The main question is whether A_{valence} differs across conditions. The directional-slope LMMs are retained as supplemental views. Starter-mechanism and Human-AI speaker-type decompositions are deferred to a follow-up pass.

Local immediate-adaptation object. We also compute an unsigned exchange-level asymmetry measure. For each exchange, the two local coupling contributions are $c_{\text{first} \rightarrow \text{second}, t} = z_{\text{predictor}} z_{\text{response}}$ and $c_{\text{second} \rightarrow \text{first}, t} = z_{\text{predictor}} z_{\text{response}}$ under the two directional pairing rules. The local asymmetry score is

$$A_{\text{local}}(t) = |c_{\text{first} \rightarrow \text{second}, t} - c_{\text{second} \rightarrow \text{first}, t}|.$$

This treats asymmetry as an exchange-level magnitude of unequal immediate adaptation, while the mixed model accounts for exchanges nested within stories.

Primary baseline-valence object. Before the alignment analysis, we also test agent-side baseline valence asymmetry. Each story contributes one unsigned baseline gap:

$$A_{\text{baseline}}(i) = |\bar{v}_{\text{slot } 1, i} - \bar{v}_{\text{slot } 2, i}|$$

where $\bar{v}_{\text{slot } 1, i}$ and $\bar{v}_{\text{slot } 2, i}$ are the mean projected valence scores for the two sides of story i . This is a baseline-level difference, not a turn-by-turn adaptation measure.

Article-use note. The primary cross-condition claim should use the unsigned per-story A_{valence} test below. Signed directional models are diagnostic only: in Human-Human and AI-AI, chronological first / second tracks starter-side ordering rather than theoretically meaningful agent identities.

1 Load Data

Data availability:

```
## # A tibble: 6 x 4
##   file                                `human-ai` `human-human` `ai-ai`
##   <chr>                                <lgl>      <lgl>      <lgl>
## 1 dyadic_sentiment_scores.parquet      TRUE      TRUE      TRUE
## 2 interaction_level_surface_metrics.parquet TRUE      TRUE      TRUE
## 3 novelty_scores.csv                   TRUE      TRUE      TRUE
## 4 semantic_exploration_binned.parquet   TRUE      TRUE      TRUE
## 5 story_embeddings_interaction_level.parquet TRUE      TRUE      TRUE
## 6 interaction_level_stories_filtered.csv TRUE      TRUE      TRUE
```

Table 1: Loaded exchange-aligned valence data by condition

condition	n_stories	n_turns	min_turn	max_turn
Human-AI	98	781	1	8
Human-Human	36	341	1	9
AI-AI	80	720	1	9

2 Primary Baseline Valence Asymmetry

This section tests whether the two sides of a story differ in their average valence baseline. The signed slot gap is only role-interpretable in Human-AI; for the cross-condition comparison, the unsigned per-story gap A_{baseline} is the main quantity.

Table 2: Per-story baseline valence asymmetry by condition

condition	n_stories	mean_A_baseline	sd_A_baseline	median_A_baseline	mean_signed_gap
Human-AI	98	0.174	0.142	0.134	-0.100
Human-Human	36	0.122	0.090	0.097	-0.004
AI-AI	80	0.084	0.063	0.075	-0.018

```
## One-way ANOVA: A_baseline ~ condition
##
##           Df Sum Sq Mean Sq F value    Pr(>F)
## condition    2  0.3586  0.17931    14.91 8.79e-07 ***
## Residuals  211  2.5379  0.01203
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Kruskal-Wallis (robust): A_baseline ~ condition
##
## Kruskal-Wallis rank sum test
##
## data:  A_baseline by condition
## Kruskal-Wallis chi-squared = 17.876, df = 2, p-value = 0.0001313
##
## Pairwise Wilcoxon (BH-adjusted):
##
## Pairwise comparisons using Wilcoxon rank sum test with continuity correction
##
## data:  df_baseline_story$A_baseline and df_baseline_story$condition
##
##           Human-AI Human-Human
## Human-Human 0.123      -
## AI-AI       9.9e-05  0.079
##
## P value adjustment method: BH
```

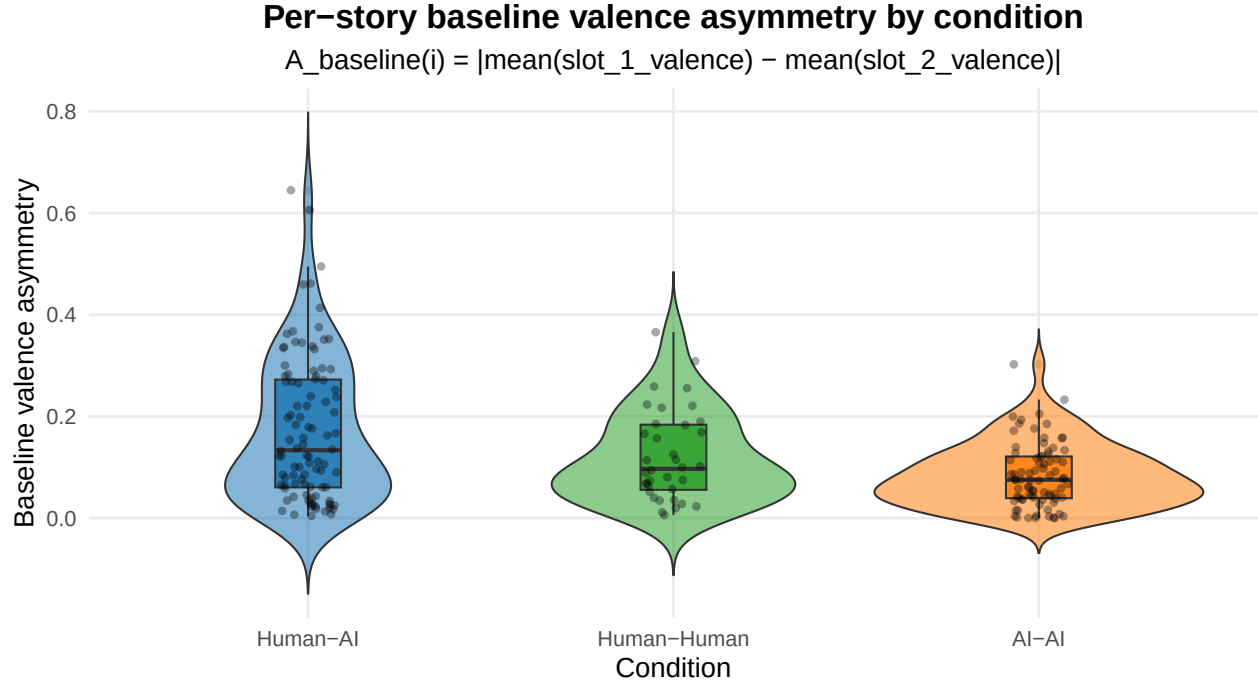


Table 3: Bootstrap 95% CI on per-condition mean of $A_baseline$ ($R = 10,000$)

condition	n	mean	ci_lower	ci_upper
Human-AI	98	0.174	0.147	0.202
Human-Human	36	0.122	0.094	0.152
AI-AI	80	0.084	0.071	0.097

Table 4: Pairwise Cliff's delta on $A_baseline$

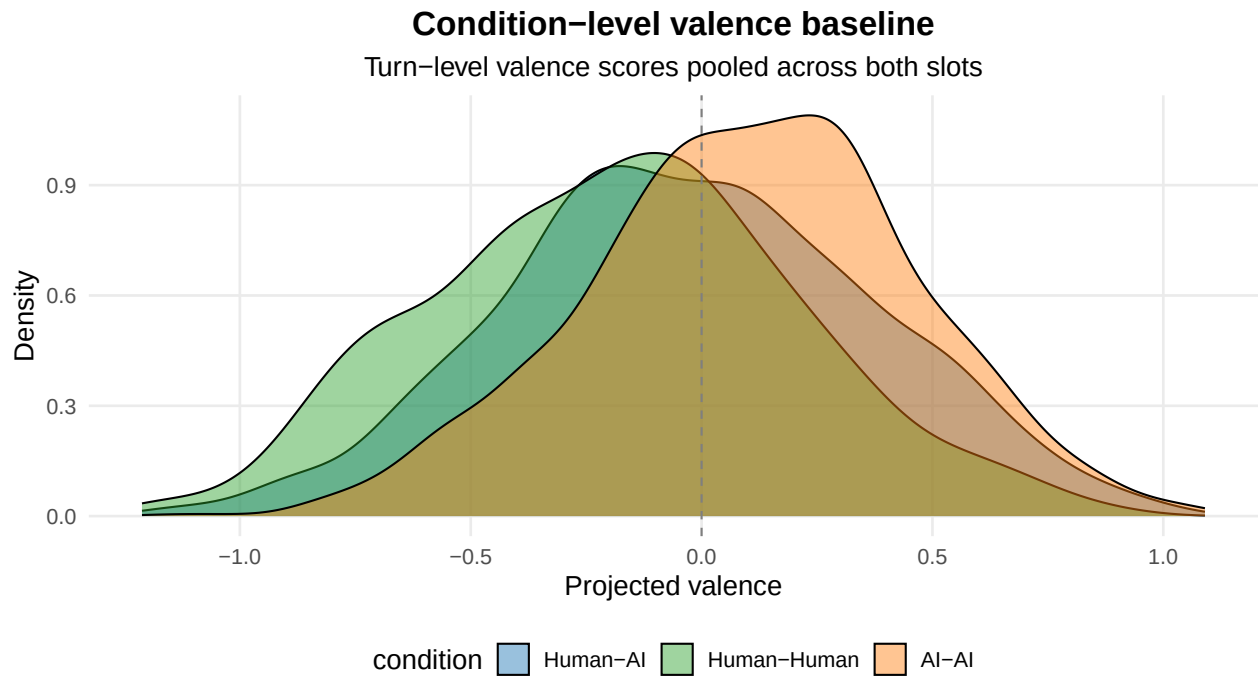
contrast	n_a	n_b	cliff_delta	magnitude
Human-AI vs Human-Human	98	36	0.175	small
Human-AI vs AI-AI	98	80	0.362	medium
Human-Human vs AI-AI	36	80	0.226	small

Table 5: Planned contrast: $A_baseline$ in HA vs the average of HH and AA

contrast	estimate	se	t	df	p	ci_lower	ci_upper
Human-AI vs (Human-Human + AI-AI)/2	0.071	0.017	4.291	142.263	0	0.038	0.104

3 Condition Baseline Valence

This plot shows overall valence baseline by condition, pooling turn-level valence scores across both slots. It is not an asymmetry measure; it asks whether the conditions differ in overall emotional tone.



4 Preprocessing

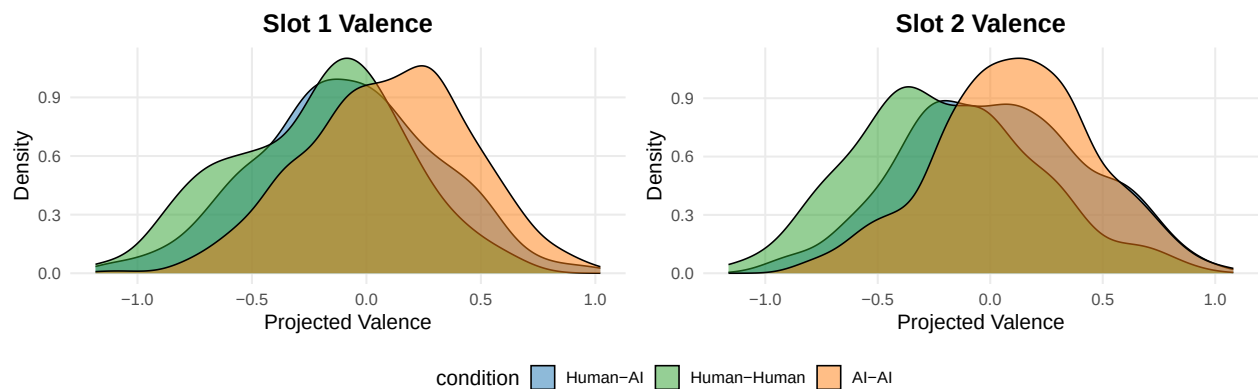
Analysis rows: 3222

Stories: 214

5 Descriptive Checks

Table 6: Mean projected valence by condition

condition	n_stories	n_turns	slot_1_mean	slot_2_mean
Human-AI	98	683	-0.088	0.023
Human-Human	36	288	-0.209	-0.203
AI-AI	80	640	0.093	0.112

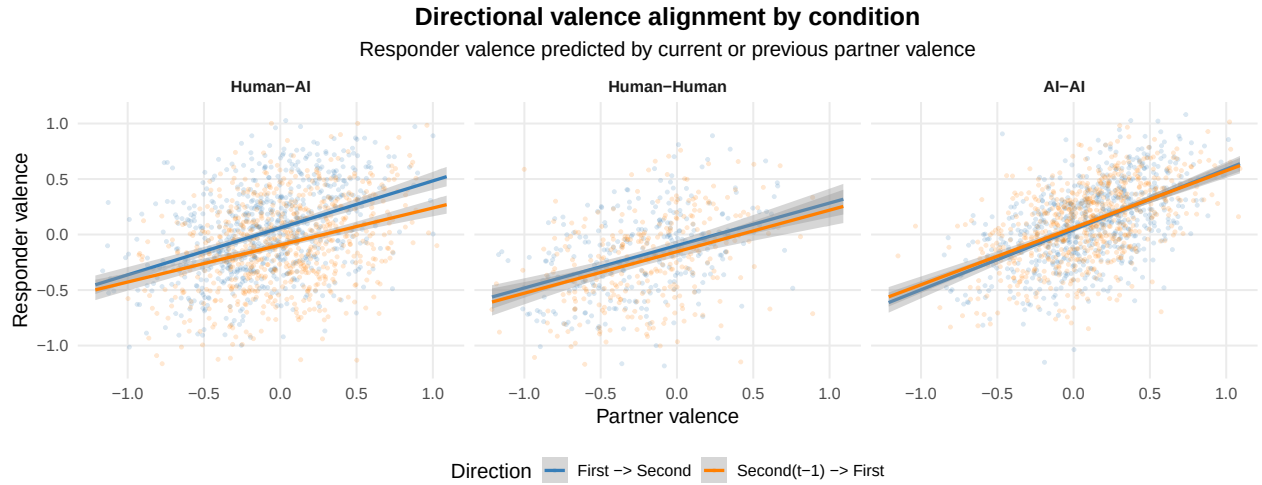


6 Directional Valence Alignment by Condition

The figure below visualizes responder valence as a function of partner valence under the two exchange-aligned pairing rules. In Human-AI, the directions map onto Human $\rightarrow$ AI and AI $\rightarrow$ Human response patterns. In Human-Human and AI-AI, they should be interpreted as chronological exchange positions rather than stable agent identities.

Table 7: Pooled descriptive slopes for responder valence predicted by partner valence

condition	alignment_direction	n_pairs	slope	intercept
Human-AI	First $\rightarrow$ Second	683	0.423	0.060
Human-AI	Second(t-1) $\rightarrow$ First	683	0.333	-0.094
Human-Human	First $\rightarrow$ Second	288	0.382	-0.099
Human-Human	Second(t-1) $\rightarrow$ First	288	0.374	-0.154
AI-AI	First $\rightarrow$ Second	640	0.543	0.046
AI-AI	Second(t-1) $\rightarrow$ First	640	0.513	0.061



7 Local Exchange-Level Alignment Asymmetry

This section treats immediate asymmetry as local to an exchange. Each directional pair contributes a standardized local coupling term, $z_{\text{predictor}}z_{\text{response}}$. The local asymmetry score is the absolute difference between the two directional coupling contributions around the same exchange index. This keeps the measure unsigned, so it does not depend on which slot or direction is larger.

Table 8: Local exchange-level alignment asymmetry by condition

condition	n_stories	n_exchanges	mean_A_local	sd_A_local	median_A_local
Human-AI	98	683	0.878	0.905	0.580
Human-Human	36	288	0.865	0.924	0.557
AI-AI	80	640	0.817	0.853	0.501

=== Local exchange-level asymmetry model ===

##	Estimate	Std. Error	df	t value	Pr(> t)
## (Intercept)	0.87770834	0.03399168	1608	25.8212706	2.790713e-123

```
## conditionHuman-Human -0.01238116 0.06241457 1608 -0.1983697 8.427809e-01
## conditionAI-AI -0.06085858 0.04887227 1608 -1.2452579 2.132184e-01
```

Table 9: Estimated marginal mean local asymmetry by condition

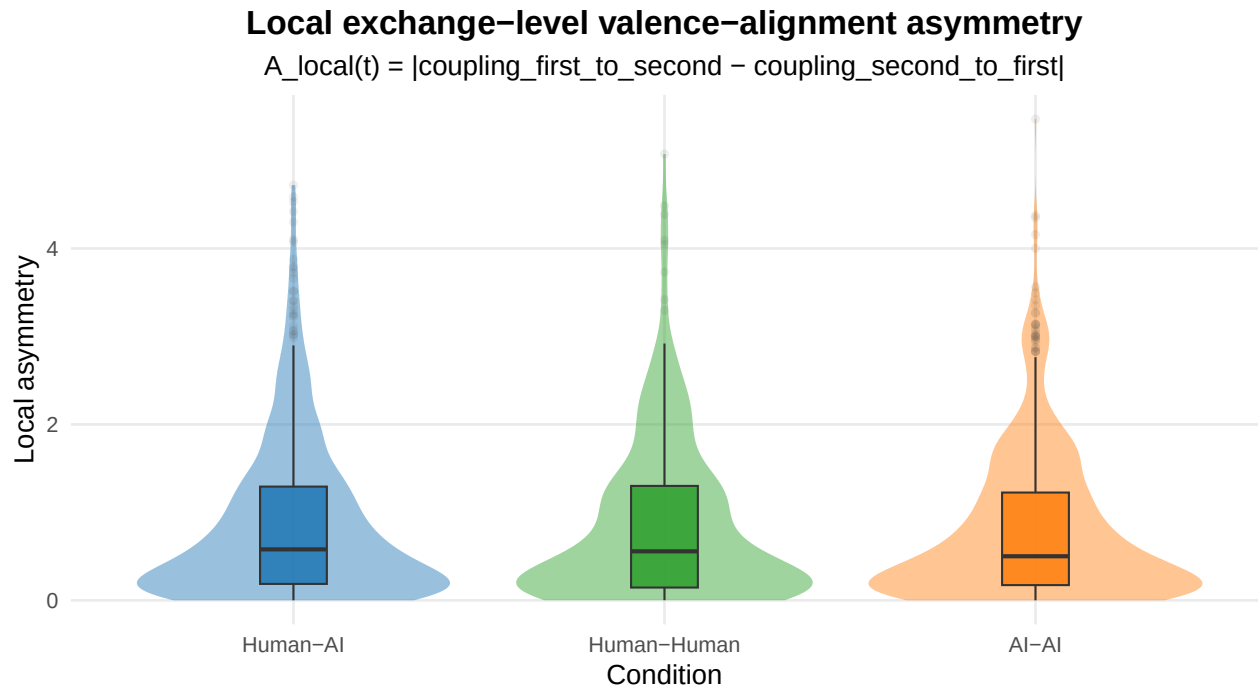
condition	emmean	SE	df	lower.CL	upper.CL
Human-AI	0.878	0.034	246.700	0.811	0.945
Human-Human	0.865	0.052	187.577	0.762	0.969
AI-AI	0.817	0.035	187.577	0.748	0.886

Table 10: Pairwise condition contrasts for local exchange-level asymmetry

contrast	estimate	SE	df	t.ratio	p.value
(Human-AI) - (Human-Human)	0.012	0.062	202.793	0.198	0.843
(Human-AI) - (AI-AI)	0.061	0.049	213.315	1.245	0.643
(Human-Human) - (AI-AI)	0.048	0.063	187.577	0.769	0.664

Table 11: Planned contrast: local exchange-level asymmetry in HA vs the average of HH and AA

contrast	estimate	SE	df	t.ratio	p.value
Human-AI vs same-type average	0.037	0.046	216.495	0.79	0.43



8 Primary: Per-Story Asymmetry Magnitude

The three-panel figure suggests that the two directional slopes may differ most strongly in Human-AI. The mixed model below tests that visual pattern directly by estimating whether the predictor-valence slope differs

by direction and whether that directional slope gap changes by condition. This is a pooled turn-level model with story-level random intercepts; the per-story A_{valence} analysis that follows is the story-level robustness layer.

```
## === Direction x condition slope-interaction model ===

##                                     Estimate Std. Error
## (Intercept)                       -4.671384e-18 0.02482269
## predictor_z                        1.354038e-01 0.02682139
## alignment_direction1               5.926804e-18 0.02482269
## conditionHuman-Human              -1.157781e-17 0.04557873
## conditionAI-AI                    2.478700e-17 0.03568936
## predictor_z:alignment_direction1    7.228943e-02 0.02682139
## predictor_z:conditionHuman-Human    9.564638e-02 0.04888141
## predictor_z:conditionAI-AI          4.363879e-02 0.03835218
## alignment_direction1:conditionHuman-Human -2.291123e-17 0.04557873
## alignment_direction1:conditionAI-AI -3.121924e-17 0.03568936
## predictor_z:alignment_direction1:conditionHuman-Human -7.964513e-02 0.04888141
## predictor_z:alignment_direction1:conditionAI-AI -3.223223e-02 0.03835218
##                                     df      t value
## (Intercept)                       3210 -1.881901e-16
## predictor_z                        3210  5.048352e+00
## alignment_direction1               3210  2.387655e-16
## conditionHuman-Human               3210 -2.540178e-16
## conditionAI-AI                     3210  6.945205e-16
## predictor_z:alignment_direction1    3210  2.695215e+00
## predictor_z:conditionHuman-Human    3210  1.956702e+00
## predictor_z:conditionAI-AI          3210  1.137844e+00
## alignment_direction1:conditionHuman-Human 3210 -5.026737e-16
## alignment_direction1:conditionAI-AI 3210 -8.747490e-16
## predictor_z:alignment_direction1:conditionHuman-Human 3210 -1.629354e+00
## predictor_z:alignment_direction1:conditionAI-AI 3210 -8.404277e-01
##                                     Pr(>|t|)
## (Intercept)                       1.000000e+00
## predictor_z                        4.704660e-07
## alignment_direction1               1.000000e+00
## conditionHuman-Human               1.000000e+00
## conditionAI-AI                     1.000000e+00
## predictor_z:alignment_direction1    7.070980e-03
## predictor_z:conditionHuman-Human    5.046905e-02
## predictor_z:conditionAI-AI          2.552706e-01
## alignment_direction1:conditionHuman-Human 1.000000e+00
## alignment_direction1:conditionAI-AI 1.000000e+00
## predictor_z:alignment_direction1:conditionHuman-Human 1.033362e-01
## predictor_z:alignment_direction1:conditionAI-AI 4.007312e-01
```

Table 12: Estimated predictor-valence slopes by condition and direction

condition	alignment_direction	predictor_z.trend	SE	df	asympt.LCL	asympt.UCL
Human-AI	First → Second	0.208	0.038	Inf	0.133	0.282
Human-Human	First → Second	0.224	0.058	Inf	0.110	0.337
AI-AI	First → Second	0.219	0.039	Inf	0.143	0.295
Human-AI	Second(t-1) → First	0.063	0.038	Inf	-0.011	0.137

condition	alignment_direction	predictor_z.trend	SE	df	asympt.LCL	asympt.UCL
Human-Human	Second(t-1) → First	0.238	0.058	Inf	0.125	0.352
AI-AI	Second(t-1) → First	0.139	0.039	Inf	0.063	0.215

Table 13: Within-condition directional slope gap

contrast	condition	estimate	SE	df	z.ratio	p.value
First_to_second_minus_second_to_first	Human-AI	0.145	0.054	Inf	2.695	0.007
First_to_second_minus_second_to_first	Human-Human	-0.015	0.082	Inf	-0.180	0.857
First_to_second_minus_second_to_first	AI-AI	0.080	0.055	Inf	1.461	0.144

Table 14: Cross-condition contrasts of the directional slope gap

contrast	estimate	SE	df	z.ratio	p.value
Human-AI vs Human-Human	0.159	0.098	Inf	1.629	0.103
Human-AI vs AI-AI	0.064	0.077	Inf	0.840	0.401
Human-AI vs same-type average	0.112	0.073	Inf	1.537	0.124

Stories with both directional correlations available:

Table 15: Per-story A_valence sample sizes by condition

condition	n
Human-AI	98
Human-Human	36
AI-AI	80

Table 16: Per-story |Fisher-z asymmetry| by condition

condition	n	mean_A	sd_A	median_A
Human-AI	98	0.612	0.471	0.554
Human-Human	36	0.516	0.405	0.439
AI-AI	80	0.490	0.355	0.425

One-way ANOVA: A_valence ~ condition

```
##           Df Sum Sq Mean Sq F value Pr(>F)
## condition    2   0.71   0.3534   2.003  0.137
## Residuals  211  37.22   0.1764
```

##

Kruskal-Wallis (robust): A_valence ~ condition

##

Kruskal-Wallis rank sum test

##

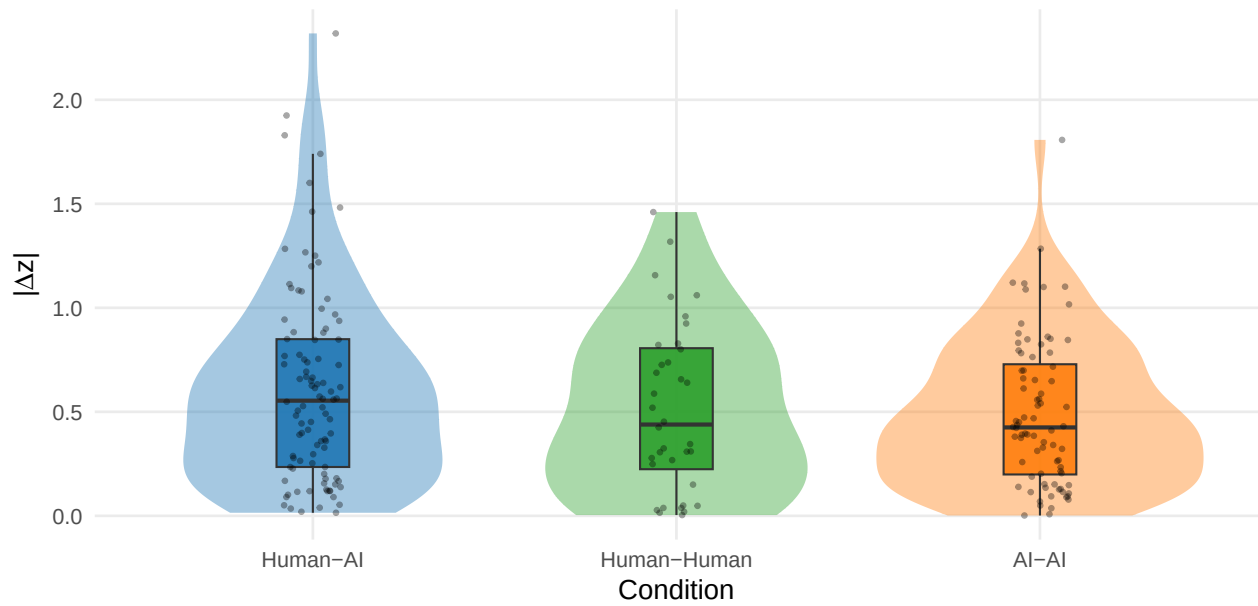
data: A_valence by condition

Kruskal-Wallis chi-squared = 2.486, df = 2, p-value = 0.2885

```
##
## Pairwise Wilcoxon (BH-adjusted):
##
## Pairwise comparisons using Wilcoxon rank sum test with continuity correction
##
## data: df_asym$A_valence and df_asym$condition
##
##           Human-AI Human-Human
## Human-Human 0.50      -
## AI-AI       0.40      0.97
##
## P value adjustment method: BH
```

Per-story valence alignment asymmetry by condition

$$A_valence(i) = |\text{atanh}(r_first_to_second) - \text{atanh}(r_second_to_first)|$$



9 Bootstrap CIs and Effect Sizes on A_valence

Table 17: Bootstrap 95% CI on per-condition mean of A_valence (R = 10,000)

condition	n	mean	ci_lower	ci_upper
Human-AI	98	0.612	0.523	0.708
Human-Human	36	0.516	0.389	0.647
AI-AI	80	0.490	0.416	0.569

Table 18: Pairwise Cliff's delta on A_valence

contrast	n_a	n_b	cliff_delta	magnitude
Human-AI vs Human-Human	98	36	0.110	negligible
Human-AI vs AI-AI	98	80	0.132	negligible

contrast	n_a	n_b	cliff_delta	magnitude
Human-Human vs AI-AI	36	80	0.004	negligible

10 Planned Contrast: HA vs (HH + AA)/2

Table 19: Planned contrast: A_valence in HA vs the average of HH and AA

contrast	estimate	se	t	df	p	ci_lower	ci_upper
Human-AI vs (Human-Human + AI-AI)/2	0.109	0.062	1.769	156.883	0.079	-0.013	0.231

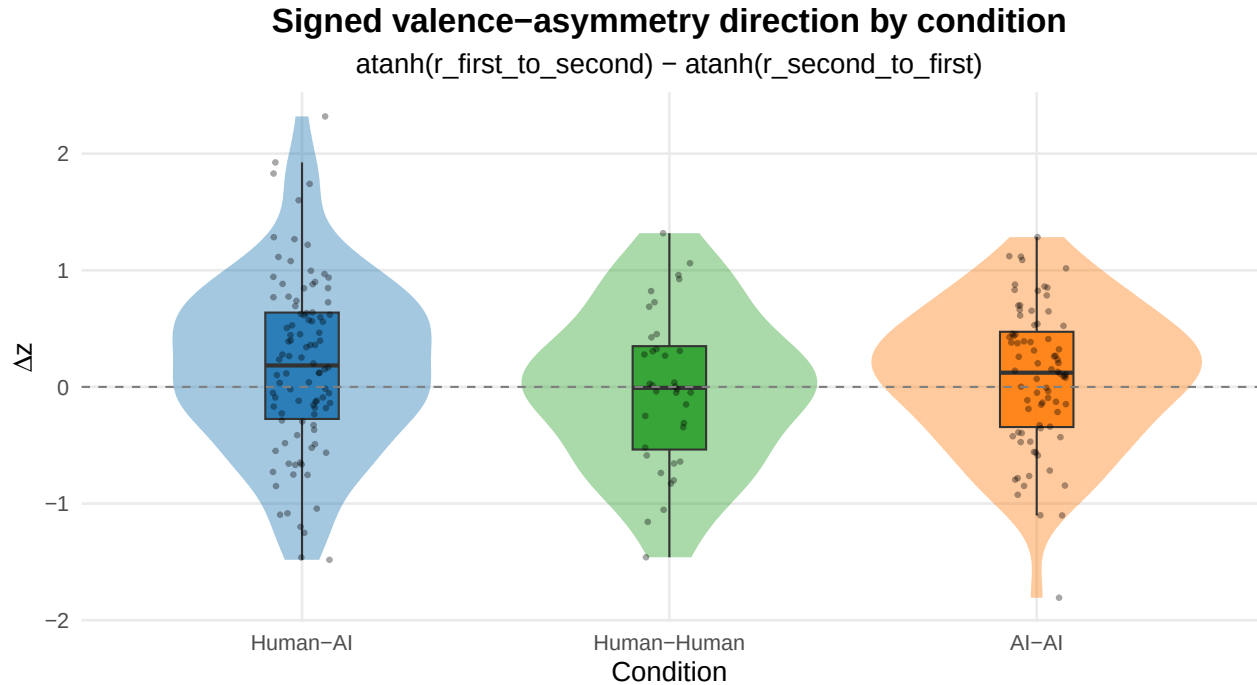
11 Directional Stability of Signed Valence Asymmetry

Tests whether the *direction* of the chronological first vs second alignment difference is consistent across stories within each condition. The signed quantity is $\text{signed}_i = \text{atanh}(r_{\text{first} \rightarrow \text{second}, i}) - \text{atanh}(r_{\text{second} \rightarrow \text{first}, i})$. HA-specific identity-tied directionality predicts high **stability** and **sign_consistency** in HA, low in HH and AA, even when unsigned A_valence is similar.

Table 20: Per-condition directional stability of signed valence asymmetry

condition	n	mean	sd	stability	sign_consistency	one_sample_t_p
Human-AI	98	0.192	0.751	0.256	0.582	0.013
Human-Human	36	-0.020	0.662	0.030	0.500	0.860
AI-AI	80	0.084	0.602	0.139	0.588	0.217

Brown-Forsythe variance-equality p = 0.1587



12 Permutation Test on $A_{\text{valence}} \sim \text{condition}$

Table 21: Permutation test on $A_{\text{valence}} \sim \text{condition}$ ($R = 10,000$ story-label permutations)

F_obs	p_perm	R
2.003	0.1407	10000

13 Summary

Primary cross-condition result: per-story $A_{\text{valence}} = |\text{atanh}(r_{\text{first} \rightarrow \text{second}, i}) - \text{atanh}(r_{\text{second} \rightarrow \text{first}, i})|$, tested with ANOVA + Kruskal and pairwise BH-adjusted Wilcoxon.

Robustness layers added:

- bootstrap 95% CIs on per-condition mean of A_{valence}
- pairwise Cliff's δ effect sizes
- planned single-df contrast HA vs (HH+AA)/2
- 10k story-label permutation test on $A_{\text{valence}} \sim \text{condition}$

The Human-AI signed Human/AI directional decomposition lives in `analysis/human-ai/valence_alignment_analysis.Rmd` and is not duplicated here.

Deferred to a follow-up pass:

- starter-mechanism analysis (signed, starter-normalized asymmetry across all conditions)

Analysis completed: 2026-05-13 10:01:08.192037